



PROJECT OVERVIEW

VirtualGlove Project Overview

Architecture, controls, security, deployment, and project status.

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Move your hand. Play the game.

VirtualGlove turns hand movement and gestures into responsive RetroArch controls using an ordinary USB camera and a **VirtualGlove Controller** built on the Arduino UNO Q. Wear a plain glove or use your bare hand-there are no sensors, wires, or electronics to add to it.

Move to steer. Curl fingers for buttons. Roll, push, pull, grab, throw, and punch. VirtualGlove recognizes the pose, sends authenticated controller input across your local network, and lets RetroArch see controller input or a native Power Glove controller.

Current stable release: v0.4.2

Development toward **v0.5.0** adds installation targets for Recalbox 10.x, Batocera 38+, and LaunchBox on Windows x86-64, while retaining RetroPie. Recalbox and Batocera create **VirtualGlove Merged Player 1**, a gamepad that combines the chosen physical controller with gestures for NES gameplay. LaunchBox adds a loopback-only RetroPad beside its physical XInput controller while retaining real keyboard controls as a manual fallback. All targets keep their services, game hooks, pairing, and configuration in each console's supported persistent storage. They are under validation and are not part of the stable v0.4.2 downloads yet. Batocera packages now carry 15 target-built, separately named Nestopia (VirtualGlove) cores for native Super Glove Ball without replacing stock Nestopia. The installer resolves the Batocera target, verifies the packaged binary and corresponding source, and load-tests the core on the console before exposing it. Recalbox packages are selected by its exact target ([rpizero2](#), [rpi3](#), [rpi4_64](#), [rpi5_64](#), [rg353x](#), [odroidgo2](#), or [x86_64](#)) and verified against their recorded ELF identity. Within Recalbox 10.x, an exact release build is preferred and the newest packaged 10.x build is the load-tested fallback. Builds are never reused across a major release. The available Raspberry Pi 3 running Recalbox's [rpizero2](#) image has passed Super Mario Bros., native Super Glove Ball, physical-joypad coexistence, and reboot-persistence checks. Recalbox 10.1 binaries and corresponding source archives are packaged for all seven targets; targets other than the tested [rpizero2](#) image remain subject to their own load and hardware validation. An unpackaged or unloadable target safely keeps using FCEUmm. LaunchBox reports conflicts affecting its manual keyboard fallback without disabling the independent VirtualGlove RetroPad. Its installer load-tests the separate Windows Nestopia core and its wrapper falls back to FCEUmm joystick mode if that DLL is later missing or changed. The v0.5.0 installers also provide a guarded upgrade from the released v0.4.2 Controller and RetroPie packages: the retired managed runtime is verified and backed up before removal, while players, calibration, tuning, pairing, device configuration, game registries, ROMs, saves, and controller assignments are preserved. Unexpected local runtime changes stop the upgrade before it writes.

Why VirtualGlove?

- **Camera-only play:** use a standard UVC camera rather than modifying a glove.
- **Two styles of NES control:** ordinary joystick output through FCEUmm and continuous native movement for Super Glove Ball through [lr-nestopia-powerglove](#).
- **Original programs and game mappings:** Programs 1-14 and A-I, including Mattel's indexed games and documented rapid-fire exceptions.
- **Fast, direct tracking:** MediaPipe Hands processes the newest camera frame and sends the latest valid hand coordinate without a settling tail.
- **Family-friendly learning:** Pixel Pal guides players through 16 Glove Academy lessons without sending accidental input to a game.
- **Personal setup:** each player can save a hand centre, movement reach, joystick centre box, gesture sensitivity, and Academy progress.
- **Game-aware and safe:** registered games select their profile automatically; stale, lost, or unauthenticated input returns to neutral.
- **Local by design:** video stays on the Controller. Pairing and controller traffic are authenticated between your own devices.

The screenshot shows the VirtualGlove Dashboard interface. At the top left is the VirtualGlove logo. To the right are navigation links: Dashboard, Play, Glove Academy, Help, Setup. Below the logo is a large heading "I love the Power Glove. It's so bad." followed by the subtitle "Live vision, gesture and controller diagnostics from your camera-only VirtualGlove." and a small robot icon. The dashboard features several status boxes: SYSTEM (Ready), PLAYER (Player One), ACTIVE PROFILE (Super Glove Ball), GAME (Startup default), HAND TRACKING (Show your hand), and CONTROLLER DELIVERY (Stopped). A "Show statistics" checkbox is present. The main area contains a camera preview placeholder with the text "Camera preview omitted" and "Documentation example — no live camera". Below this are four buttons: Center hand, Start controller, Connection, and Shutdown. On the right, the "CURRENT PROGRAM" section displays "Super Glove Ball" with a description: "Continuous Robo-Glove control with a joystick fallback." and a paragraph explaining the custom Nestopia core. Below this is a table of controls:

Move hand	Continuous X/Y or D-pad
Curl index	A
Curl thumb	B
Close hand	Grab / close Robo-Glove
Point index	Robo-Bullet

At the bottom of the program section is a link: [Open the Gameplay Guide](#). The footer of the dashboard shows "VirtualGlove v0.4.0-dev" and "► Software and matrix firmware".

VirtualGlove Dashboard with camera and controller status

What you need

- An Arduino UNO Q provisioned through Arduino App Lab
- A supported console: RetroPie, Recalbox 10.x, Batocera 38+, or LaunchBox with 64-bit Windows and 64-bit RetroArch (the latter three are v0.5.0 targets)
- A UVC-compatible USB camera and powered USB hub
- A physical controller for RetroArch setup and recovery
- Both devices on the same trusted local network with internet access during installation
- Your own legally obtained games-VirtualGlove includes no ROMs or BIOS files

New to the hardware? Start with [Build your own](#) for the parts, expected cost, and difficulty.

Install VirtualGlove

These steps install the current stable release. Install the **same version on both devices** and close any running RetroArch game first. The scripts verify their downloads, ask for administrator access when needed, and preserve existing pairing and player settings during an update.

1. Prepare the Controller

Finish the UNO Q's App Lab setup, connect it to your network, and attach the camera through the powered hub. The camera may also be connected after installation.

2. Install the Controller software

Open a terminal on the UNO Q and run:

```
SH
cd /home/arduino
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/download/v0.4.2/install-uno-q.sh
bash install-uno-q.sh --version v0.4.2
```

On a first installation, the installer suggests **virtualglove** as the Controller name, making its usual address virtualglove.local. Press Enter to accept it or type a different family-friendly name. Updates preserve the existing name.

3. Check the Controller

Open the Dashboard address printed by the installer-normally <http://virtualglove.local:8088/dashboard>. The installer configures automatic startup, the matrix display, guarded camera recovery, and its required host helpers.

4. Install the console software

For the current stable v0.4.2 release, open a terminal on RetroPie and run:

```
SH
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/download/v0.4.2/install-retropie.sh
bash install-retropie.sh --version v0.4.2
```

If an older Buster-based RetroPie reports that its Raspbian repository has no Release file, stop and follow [Buster package source moved](#), then rerun this step. The installer does not silently rewrite operating-system repositories.

For v0.5.0, choose exactly one console installer:

Console	Run on the console	Installed in
RetroPie	<code>bash install-retropie.sh --version VERSION</code>	<code>/opt/virtualglove-src</code> and <code>/etc/virtualglove</code>
Recalbox	<code>bash install-recalbox.sh --version VERSION as root</code>	<code>/recalbox/share/system/virtualglove</code>
Batocera	<code>bash install-batocera.sh --version VERSION as root</code>	<code>/userdata/system/virtualglove</code>
LaunchBox	<code>install-launchbox.ps1</code> as the Windows player	<code>%LOCALAPPDATA%\VirtualGlove</code>

Replace `VERSION` with the same published v0.5.0 prerelease or release tag used on the Controller. Download the matching Linux installer from that release's GitHub assets. LaunchBox is installed from the extracted Windows package. The complete [Installation Guide](#) has copyable commands for all four platforms.

Each installer uses the platform's persistent storage, preserves ROMs, saves, the game registry, pairing, and unrelated controller configuration, and keeps a physical Player 1 joypad usable beside VirtualGlove. Batocera automatically selects the native core only for exact registered Super Glove Ball filenames that do not already have an explicit core choice. A target without a loadable native core still receives the complete FCEUmm joystick path.

The generic RetroPie installation exposes a separate **VirtualGlove** gamepad; the cabinet's pre-existing combined-Player-1 merger remains a specialized local configuration. Recalbox and Batocera instead ask which configured controller is Player 1 and expose one **VirtualGlove Merged Player 1** gamepad to NES RetroArch. The original controller remains the only active frontend controller. LaunchBox keeps physical XInput plus a loopback Network RetroPad for FCEUmm games; its real keyboard mappings remain available as a manual fallback. Native Super Glove Ball uses only the guarded Power Glove state channel; it does not duplicate native gestures as keyboard events. Its installer makes **VirtualGlove RetroArch** the default NES emulator after backing up LaunchBox's emulator and NES game definitions. Existing NES games assigned to standard RetroArch and new NES imports use VirtualGlove automatically; games assigned to a genuinely different emulator remain explicit overrides.

5. Pair the devices

Open the secure Setup address printed by the installer—normally <https://virtualglove.local:8443/setup>—open **Connection and startup**, choose RetroPie, Recalbox, Batocera, or LaunchBox, save the console hostname or IP address, and continue into **Pair this Controller**. Pairing stays unavailable until the platform and address are both saved. The guided one-time-code method is recommended and shows only the command for the selected platform. For an optional, resumable first-game walkthrough, select **Get ready to play** on Setup or Dashboard. It checks your player, connection, camera, center, and essential gestures in safe practice before you explicitly enable game controls.

After confirming the browser certificate against the physical Matrix ID, the optional **Trust this Controller** step removes future privacy warnings on that phone or computer.

6. Set up a player

Choose a player, position the camera, and use **Center hand**. Open **Glove Academy** to learn the gestures and adjust movement reach or sensitivity only if needed.

7. Play

Select **Start controller**, launch a registered game, and confirm the expected profile. FCEUmm uses joystick mode; Super Glove Ball can also use the optional native core selected per ROM on RetroPie, Recalbox, Batocera, or LaunchBox.

ROMs added later do not require reinstalling VirtualGlove. Refresh the console frontend's game list, then open **Setup -> Games** and register the exact ROM filename if it is not already one of the supplied aliases. RetroPie uses its per-ROM launch choice for native Super Glove Ball; Recalbox and Batocera apply their exact-ROM native choice at VirtualGlove service startup; LaunchBox's VirtualGlove RetroArch wrapper chooses the core on every launch. The complete installation guide lists the platform-specific refresh and restart steps.

The complete [Installation Guide](#) has first-install checkpoints, illustrated pairing, camera advice, native-core setup, updates, backups, and troubleshooting. Use it as the authoritative setup reference.

What can you play?

VirtualGlove includes the original Programs 1-14, nine reusable cartridge Programs A-I, plus dedicated mappings for Bad Street Brawler and Super Glove Ball. The same recognition settings follow the player across games; profiles change only what the recognized movements and gestures send to the console.

Path	What it provides
FCEUmm	The selected Programs 1-14 or A-I mapping, including positional movement, mapped A/B actions, and documented compound gestures.
Super Glove Ball with FCEUmm	A complete joystick-mode fallback that can always be selected for testing or play.
Super Glove Ball with Nestopia (VirtualGlove)	Continuous native X/Y and Z, Start, grab/catch, release/throw, Robo-Bullet fire, and Power Punch. The isolated core is supported on RetroPie, Recalbox, Batocera, and LaunchBox; FCEUmm remains the fallback.

Rapid A/B defaults on only where an individual program description explicitly identifies a pulsed button: Program 7 A, Program B A, Program H A/B, and Bad Street Brawler B. Every other profile defaults off. The registered Blaster Master, Double Dribble, Racket Attack, Ice Hockey, and Alpha Mission entries retain Mattel's documented off instructions. Programs 13 and 14 also support deliberate physical-controller and gestures-off play.

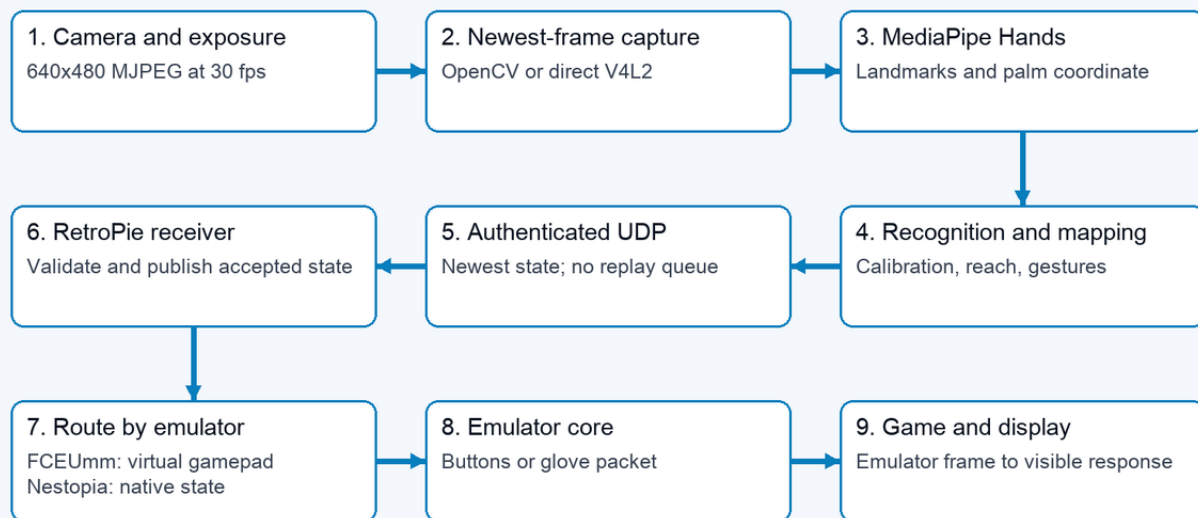
Dashboard Rapid A/B switches control button repetition only; profile-owned fast turns, pulsed movement, turbo movement, and compound actions retain their own timing. Saved per-game overrides survive upgrades. Choose **Use profile defaults** while that game is running to remove its overrides and adopt the corrected defaults.

The [Gameplay Guide](#) shows every numeric Program gesture, compound action, release rule, indexed game, exception, objective, and practice challenge. The [Native Emulation guide](#) explains why the two emulator paths feel different.

The numeric Program section includes gesture-by-gesture controls, compound action timing, the complete official game index, and the five automatic rapid-fire exceptions. Program 13 is the mixed physical-controller option; Program 14 deliberately turns camera gestures off for the active game.

How it works

08 / Camera to game: one front end, two delivery paths



The browser preview and statistics observe the worker; neither belongs in the controller delivery path.

End-to-end VirtualGlove flow from camera to game

1. The camera delivers its newest frame to the VirtualGlove Controller.
2. MediaPipe Hands finds the palm, wrist, and finger landmarks.
3. Shared recognition turns those landmarks into position, fingers, rolls, depth motion, and menu poses.
4. The Controller sends the newest authenticated state to the selected console.
5. The console publishes either Player 1 input or native glove state for the selected emulator core.

Pairing is tied to a private shared key rather than one permanent IP address. If DHCP changes an address or `.local` resolution temporarily fails, the paired devices can rediscover one another on the local network without broadcasting controller states or requiring a new pairing.

Meet Pixel Pal

Pixel Pal helps players learn, personalize, test, and troubleshoot without turning setup into an engineering exercise.

- **Glove Academy** teaches all 16 movements and gestures while game output is paused.
- **Tune gestures** records guided examples and previews a conservative adjustment before saving it.
- **Find the best camera settings** compares only choices supported by the attached camera and changes nothing until the player accepts a recommendation.
- **Rock Paper Scissors** provides a camera-controlled practice game that does not require a connected console.

Controller pages

Page	Purpose
Dashboard · /dashboard	See live controls and set per-game A/B rapid fire.
Play · /play	Challenge Pixel Pal to Rock Paper Scissors.
Glove Academy · /learn	Learn gestures, set movement reach, and personalize recognition safely.
Setup · /setup	Manage players, camera choices, console pairing, games, backups, and display preferences.
Help · /help	Read the complete manuals and printable PDFs directly on the Controller.

Documentation

Start here

You want to...	Read...
Install, pair, and play your first game	Installation Guide
Learn gestures, Programs, and game controls	Gameplay Guide
Choose or troubleshoot a camera	Camera Guide
Recognize matrix animations and messages	Matrix Display Guide
Find a quick command or status reminder	Quick Reference
Solve a problem by symptom	Troubleshooting

Go deeper

You want to...	Read...
See the current components and data flow	Architecture
Understand joystick and native emulation	Native Emulation Explained
Review proven Super Glove Ball behavior	Native Compatibility Record
Look up every setting and command	Configuration Reference
Follow the one-week engineering process and experiments	Engineering Journey
Review security and pairing boundaries	Security Policy
Check dependencies and third-party terms	Third-party Notices
Contribute code or documentation	Contributing Guide

Printable editions of all maintained guides are available in [output/pdf/](#). The Controller serves the same documentation from its local Help page.

Project status

The stable v0.4.2 release keeps the proven CPU MediaPipe Hands path, Programs 1-14, guided readiness checks, and live dead-zone visualization. It adds live rapid-fire controls, source-accurate rapid defaults, leaner documentation and deployment packages, cached Dashboard status inputs, and removal of retired pre-0.4.1 compatibility and protocol paths. The release updates the UNO Q and RetroPie together while preserving pairing, players, calibration, tuning, Academy progress, and the installed game registry.

The v0.5.0 development line extends the same authenticated design to Recalbox, Batocera, and LaunchBox. Recalbox and Batocera use a persistent merged Player 1 gamepad with a separately selected frontend controller; LaunchBox uses a LAN-isolated local RetroPad while leaving XInput and real keys available. Recalbox on the available Raspberry Pi 3 has passed registered FCEUmm play, native Super Glove Ball, simultaneous physical-joypad use, and reboot persistence. Batocera's 15 packaged architectures and LaunchBox's Windows x86-64 DLL have reproducible build, package, and runtime checks; live Batocera and LaunchBox gameplay acceptance remains required before release. Different cameras, rooms, players, controllers, operating-system images, and console architectures remain valuable real-world tests.

Use Setup's **Download system report** when asking for help. It records useful software, camera, controller, and connection health without including video, pairing keys, player calibration, ROM names, or network addresses.

Contributing and licensing

Issues, careful test reports, documentation improvements, and code contributions are welcome. Please read the [Contributing Guide](#) before opening a change. Release history is kept in the [Changelog](#).

VirtualGlove is maintained by **Iain Bennett** and is licensed under the [MIT License](#). The modified Nestopia core is GPLv2 software and remains separate from the MIT application. Nintendo, NES, Power Glove, and the named games belong to their respective owners. See [Third-party Notices](#) for dependency, model, asset, and native-core licensing details.